# LAVANYA SENTHIL

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## EDUCATIONS

| Year | Degree | Institute | CGPA/MARKS |
|---|---|---|---|
| Aug 2023 - 2027 | B.Tech in CSE (AIML) | PES University | 8.58 / 10 |
| Jun 2021 - 2023 | XII, CBSE | Delhi Public School | 91.2% |
| Jun 2009 - Jun 2021 | X, ICSE | Baldwin Girls' High School | 97.6% |

## WORK EXPERIENCES

**AI/ML Intern** | Internship — Jun 2026 - Jul 2026
*AVEVA R&D — E3D Team* — *On-site • Hyderabad*

- Designed a repository-aware multi-agent AI system to auto-generate compilable C#/.NET unit tests for enterprise-scale codebases.
- Adopted a **6-agent architecture** with isolated context per agent, **improving output consistency from 20% to 35% coverage** across the pipeline.

***Technologies / Skills Used :*** *C#, .NET, Agents*

**AI/ML Research Intern** | Internship — Jun 2025 - Aug 2025
*C3I (ML Lab), PES University* — *On-site • Bangalore*

- **Designed and implemented** a deep learning system for visual detection of spice adulteration using **CNNs**.
- **Improved classification accuracy** through preprocessing and model optimization techniques.

***Technologies / Skills Used :*** *CNN, Resnet, DeepLearning, FastApi*

## PROJECTS

**Multi-Cloud Behavioral Anomaly Detection Platform (Capstone- Ongoing)** — Aug 2025 - Present

- Built a real-time platform to detect and prioritize anomalous behavior across 50,000+ cloud infrastructure metrics, helping IT teams focus on critical alerts and identify root causes for faster resolution.

***Technologies / Tools Used :*** *Python, PyTorch, Kafka, FAISS, Redis, Neo4j, FastAPI, Self-Supervised ML, AWS*

**AI-Based Visual Detection of Spice Adulteration** — Jun 2025 - Aug 2025

- Developed a **CNN-based system** to detect spice adulteration from a single smartphone image, enabling users to assess adulteration without specialized equipment; achieved **~99% F1 score** on **10,000+ images**.

***Technologies / Tools Used :*** *ML, CNN, Transfer Learning, Preprocessing, OpenCV, FastApi*

**Drug-Likeness Prediction using Feature-Enriched GNNs** — Jan 2025 - May 2025

- Built a GNN to predict drug-likeness by combining molecular structure and global chemical properties, helping prioritize promising drug candidates for early-stage screening; achieved 0.927 ROC-AUC and 0.935 F1 with a 2.6% gain over a structure-only model.

***Technologies / Tools Used :*** *Python, PyTorch Geometric, RDKit*

**Self-Correcting Text-to-Image Generation via Scene Graph-Grounded VQA** — Feb 2026 - Apr 2026

- Built a self-correcting text-to-image system to reduce semantic mismatches between prompts and generated images using scene graphs and VQA.

***Technologies / Tools Used :*** *LLMs, VLMs*

## SKILLS

| | |
|---|---|
| **Databases :** | MySQL, MongoDB, Vector Databases, Neo4j |
| **Frameworks & Libraries :** | TensorFlow, Keras, scikit-learn, NumPy, pandas, PyTorch |
| **Programming Languages :** | Python, C, Java, JavaScript |
| **Soft Skills :** | Communication, Problem Solving |
| **Tools & Platforms :** | Git, GitHub, Jira, Docker, AWS, CI/CD |

## AWARDS & ACHIEVEMENTS

- **CNR Scholarship** — Awarded to the **top 20% of students**; 4× recipient
- **1st Prize — Kepler's Habitability Datathon** — Won first place in a data science competition focused on planetary habitability
- **JEE Main** — **93.3 percentile** in the national-level engineering entrance examination
- **Academic Excellence Award** — Baldwin Girls' High School; **97.6%** in Class X Boards